Internal Dimensional Volume, Speciation, and the Marble Intuition in CRF

IDV as Complexity Measure, Not Crystallisation Measure

Simulation results from CRF Hybrid V17–V18

Ougit Jarittum

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Abstract

This paper introduces and tests Internal Dimensional Volume (IDV), a node-level measure of accessible state richness in CRF simulations. IDV is grounded in the δ-chain: $V_i^{\text{int}} = H(P_i) \cdot (1 + \phi\text{-diversity}) \cdot (1 + |\nabla\Psi_i|) \cdot \ln(1 + \text{mass}_i)$, where each factor measures a distinct dimension of P-space complexity.

Three tests were conducted in V18:

Test 1 (Speciation): The pattern-mode field Ψ_i separates into 5 distinct modes (histogram peaks) by sweep 400. This confirms that selective coupling $e^{-|\Psi_i - \Psi_j|/\sigma}$ produces genuine matter-type differentiation.

Test 2 (Marble Effect): High-IDV nodes (Group X) show $d_s = 4.35$ at late times, while low-IDV nodes (Group Y) show $d_s = 1.92$. This is the opposite of the prediction that IDV pulls $d_s \to 3$. The finding reveals a fundamental distinction: IDV measures *complexity* (open, fluctuating, high-entropy nodes), not *crystallisation* (converged, stable, low-entropy nodes). Nodes approaching $d_s \to 3$ are *losing* IDV, not gaining it. This reframes IDV as a measure of a node's *informational youth*.

Test 3 (Gravity): $R^2 = 0.908$, slope $= 1.24$ — gravity is preserved through all CWRC extensions.

The paper concludes with a revised definition of IDV, an honest assessment of what the marble intuition really means in CRF, and the open question of whether IDV and Ω (the crystallised mass field) are two faces of the same δ-chain.

Contents

1 IDV: Definition and δ-Chain Grounding

Internal Dimensional Volume

For node i with k neighbours $\mathcal{N}(i)$:

$$V_i^{\text{int}} = H(P_i) \cdot \big(1 + \text{Var}_{j\in\mathcal{N}(i)}[\cos(\Phi_j - \Phi_i)]\big) \cdot \big(1 + \overline{|\Psi_j - \Psi_i|}_{j\in\mathcal{N}(i)}\big) \cdot \ln(1 + \text{mass}_i)$$

Local effective dimension: $d_i^{\text{int}} = \ln(1 + V_i^{\text{int}})$.

Each factor maps to a CRF quantity:

IDV factor	**CRF origin**
$H(P_i)$	Entropy of $C_i \equiv \Pr(s\|C)$. From HDynamics: $H \to 0$ as C crystallises.
ϕ-diversity	Phase coherence of DKL wave field. Low diversity = node near Axiom 11 attractor ($G \to 0$).
$\|\nabla\Psi\|$	Speciation boundary sharpness. High gradient = node at the edge between matter types.
$\ln(1 + \text{mass})$	$\ln(1 + \Omega_i/\kappa)$, log of accumulated R-event history.

Status: **Hypothesis** — product form not yet derived from axioms. Each factor is individually grounded; their independence is assumed.

2 Test 1: Speciation Confirmed

Speciation: 5 Psi Modes Detected

V18 configuration: $\sigma = 0.3$ (strict selective coupling), $N = 500$, 400 sweeps.
Result: the Ψ_i histogram at sweep 400 shows 5 distinct peaks (identified by requiring height $> 20\%$ of maximum, spacing > 3 bins).
Interpretation: selective coupling $\kappa_{ij} = D_{\text{KL}}(P_i \| P_j) \cdot e^{-|\Psi_i - \Psi_j|/\sigma}$ causes nodes with similar Ψ to reinforce each other and diverge from dissimilar nodes. This is *speciation* in CRF language: clusters of nodes that share low pairwise D_{KL} and similar Ψ constitute distinct *matter types*.
In CRF: each species corresponds to a cluster in P_{shared} with high $\Pr(s)$ for a specific pattern s — the same mechanism as CRF Synchronicity (collective dynamics).

3 Test 2: The Marble Effect — An Unexpected Finding

3.1 Prediction and Result

The original prediction was: high-IDV nodes have stronger 3D metric coupling $\Rightarrow$ their random walkers are trapped in 3D $\Rightarrow d_s^X < d_s^Y$.

The simulation found the opposite:

Group	**Mean IDV**	d_s **(late)**
X (top 20% IDV)	high	4.35
Y (bottom 20% IDV)	low	1.92

High-IDV nodes have *higher* d_s, not lower.

3.2 What This Means

IDV = Informational Youth, Not Crystallisation

High IDV requires high $H(P_i)$ (entropy). High entropy means C_i is spread: the node has many accessible patterns, has not yet converged, is still exploring.
From HDynamics: $H \to 0$ as the node approaches its attractor. Nodes with $d_s \to 3$ are nodes whose random walkers are restricted to a 3D subspace — these are *crystallised* nodes, which by definition have low H and therefore *low IDV*.
The Marble Effect is real, but inverted:

- High IDV $\Leftrightarrow$ node is informationally young, complex, fluctuating
- Low IDV $\Leftrightarrow$ node is crystallised, stable, approaching Axiom 11
- $d_s \to 3$ happens as IDV *falls*, not as it rises

The marble intuition survives — but "more space inside" means more accessible patterns (youth), not more geometric dimensions. As a marble ages (gains mass, converges), its internal space *contracts* toward a stable 3D geometry.

3.3 Revised Interpretation of the Marble Intuition

The original intuition: two marbles of the same external size can have radically different internal volume.

The CRF interpretation after V18: a young node (high IDV) is like a marble full of possibility — many patterns accessible, geometry still forming. An old node (low IDV, high Ω) is like a marble that has crystallised — few patterns remain accessible, geometry is fixed at 3D.

Two nodes at the same X-position can have different IDV because they have different histories. The older one has resolved more distinctions, compressed its constraint distribution, and locked its geometry. The younger one is still open.

IDV–Ω Duality (Derived)

From the δ-chain:

$$\mathrm{IDV}_i \uparrow \iff H(C_i) \uparrow \iff \Omega_i = D_{\mathrm{KL}}(C_i \| P_{\mathrm{bg}}) \downarrow$$

$$\Omega_i \uparrow \iff H(C_i) \downarrow \iff \mathrm{IDV}_i \downarrow$$

IDV and Ω are anti-correlated across the node lifecycle. A node begins with high IDV and low Ω (vacuum, uncrystallised). As R-events accumulate: $\Omega \uparrow$, $H \downarrow$, IDV$\downarrow$. The node becomes matter — concentrated, stable, gravitationally active.
The δ-chain describes this as a one-way process: $\delta \to R$-events $\to C \uparrow \to H \downarrow \to$ crystallisation. IDV tracks the *remaining openness* at each stage.

4 Test 3: Gravity Preserved

Gravity Unaffected by CWRC Extensions

With Φ, Ψ, and IDV active in V18:

$$R^2 = 0.908, \quad \text{slope} = 1.24, \quad G_{\mathrm{sim}}/\ln 2 = 0.202$$

Identical within error to V16 (no wave fields): $R^2 = 0.906$, slope $= 1.24$, $G_{\mathrm{sim}}/\ln 2 = 0.197$.
Conclusion: the Ω-field gravity mechanism is orthogonal to the wave/identity dynamics. $\Omega_i = \mathrm{mass}_i - 1$ encodes only the R-event history, which is not modified by Φ or Ψ evolution. This confirms that the two papers (v5 and this one) describe *different layers* of the same δ-chain, not competing mechanisms.

5 The Two Layers of CRF Dynamics

V16–V18 together reveal a two-layer structure:

Layer	**Field**	**CRF quantity**
Crystallisation	$\Omega_i = \mathrm{mass} - 1$	$D_{\mathrm{KL}}(C_i \| P_{\mathrm{bg}})$, gravity, $d_s \to 3$
Complexity	IDV_i	$H(P_i)$, speciation, informational youth
Anti-correlated: $\Omega \uparrow \Leftrightarrow$ IDV $\downarrow$		
Both derive from $\delta \to R$-events $\to C$		

Gravity (Paper v5) is the crystallisation layer. This paper is the complexity layer. The full node lifecycle runs from IDV-dominated (young, open) to Ω-dominated (old, crystallised) as $t = |R_{\text{events}}|$ increases.

6 Status Table

Result	**Status**	**Note**
Speciation: 5 Ψ modes	Confirmed	V18 simulation
Gravity preserved through V18	Confirmed	$R^2 = 0.908$
IDV $\uparrow$ as $\Omega \downarrow$	Derived	From δ-chain
Marble = informational youth	Derived	$H \uparrow \Leftrightarrow$ IDV$\uparrow$
Marble Effect (IDV $\to d_s \to 3$)	Falsified	Opposite found
IDI stabilises d_s variance	Open	Corr = +0.18, not negative
IDV product form independence	Open	Not yet proved

The marble full of space is the young one.
The marble that has crystallised has less space inside —
but that space is 3-dimensional, stable, and gravitationally alive.
Youth is richness. Age is precision.
Both follow from δ.